Centre Number	Candidate Number	Name

UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education Ordinary Level

CHEMISTRY **5070/02**

Paper 2 Theory

May/June 2004

1 hour 30 minutes

Candidates answer on the Question Paper.
Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your Centre number, candidate number and name on all the work you hand in.
Write in blue or black pen.
Do not use staples, paper clips, highlighters, glue or correction fluid.
You may use a calculator.

Sections A
Answer **all** questions.
Write your answers in the spaces provided on the Question Paper.

Section B
Answer any **three** questions.
Write your answers on any lined pages and/or separate answer paper.
At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.
A copy of the Periodic Table is printed on page 16.

If you have been given a label, look at the details. If any details are incorrect or missing, please fill in your correct details in the space given at the top of this page.

Stick your personal label here, if provided.

For Examiner's Use	
Section A	
B7	
B8	
B9	
B10	
Total	

This document consists of **15** printed pages and **1** lined page.

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Section A

Answer **all** the questions in this section in the spaces provided.

A1 Choose from the following substances to answer the questions below.

argon

calcium phosphate

ethene

lead(II) nitrate

methane

phosphorus oxide

potassium nitrate

sulphur dioxide

Each substance can be used once, more than once, or not at all.

Name a substance which,

(a) is a greenhouse gas produced by the decay of vegetable matter,

.....[1]

(b) contains **two** of the essential elements needed by plants,

.....[1]

(c) reacts with warm aqueous sodium hydroxide and aluminium powder to form a gas that turns moist red litmus blue,

.....[1]

(d) dissolves in water to form a solution which neutralises sodium hydroxide.

.....[1]

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A2 Two isotopes of potassium are $^{39}_{19}K$ and $^{40}_{19}K$.

(a) Complete the table about the number of particles found in one atom of each of these isotopes.

	protons	number of electrons	neutrons
$^{39}_{19}K$			
$^{40}_{19}K$			

[2]

(b) Potassium reacts with water as shown in the equation.

$$2K(s) + 2H_2O(l) \longrightarrow 2K^+(aq) + 2OH^-(aq) + H_2(g)$$

Describe what you would see when potassium reacts with water.

.....

.....

.....[2]

(c) A sample of 0.195 g of potassium was added to 500 cm^3 of cold water. When the reaction was finished, 100 cm^3 of 0.100 mol/dm^3 hydrochloric acid was added to form solution **X**.

(i) Calculate the number of moles of hydroxide ions formed when the potassium was added to water.

(ii) Calculate the number of moles of hydrogen ions in 100 cm^3 of 0.100 mol/dm^3 hydrochloric acid.

(iii) Give an ionic equation to represent the neutralisation reaction.

.....

(iv) Suggest a pH value for solution **X**.
Explain your answer.

.....

.....
[4]

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[Turn over

(d) Potassium oxide is an ionic solid.

Draw the electronic structure of both a potassium ion and an oxide ion.
Include the charge on each ion.

Potassium ion

Oxide ion

[2]

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A3 More than 60 000 plastic materials, or polymers, are in use.

The table gives some information about five important polymers.

polymer	density in kg/m^3	maximum useable temperature / °C	solubility in organic solvents
low density poly(ethene)	920	85	soluble above 80 °C
high density poly(ethene)	960	120	soluble above 80 °C
poly(phenylethene)	1050	65	soluble
poly(chloroethene)	1390	60	soluble
poly(propene)	900	150	insoluble

(a) Which polymer would be most suited for making a pipe to carry lubricating oil at 100 °C? Give **two** reasons for your answer.

answer .....

reasons .....

.....

.....[2]

(b) State **one** use for poly(ethene).

.....[1]

(c) Describe some of the problems of the disposal of waste polymers.

.....

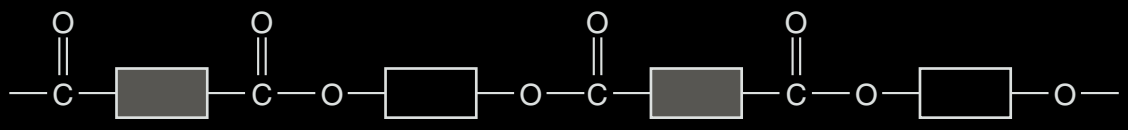
.....

.....[2]

(d) Poly(propene) is made from the monomer propene.
Draw the structure of poly(propene).

[2]

(e) *Terylene* is a condensation polymer.
The structure of *Terylene* is shown below.



(i) What is the name of the linkage shown in the structure of *Terylene*?

.....

(ii) Name a natural macromolecule that contains the same linkage as *Terylene*.

.....

[2]

(f) Draw the structure of a polyamide such as nylon.

[1]

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A4 The exhaust fumes from the internal combustion engines of motor vehicles contribute to the poor quality of air in many cities. The exhaust fumes contain atmospheric pollutants such as nitric oxide, NO, and carbon monoxide, CO.

(a) Nitric oxide, NO, is formed when oxygen and nitrogen from the air react in an internal combustion engine.

(i) Construct a balanced equation for this reaction.

.....

(ii) Explain why, in terms of collisions between particles, the rate of this reaction increases as the concentration of oxygen increases.

.....

.....

(iii) Explain why the rate of this reaction increases as the engine temperature increases.

.....

.....

[4]

(b) Explain how carbon monoxide is formed in an internal combustion engine.

.....

.....[1]

(c) Nitric oxide and carbon monoxide in the exhaust gases react together in the catalytic converter of a motor vehicle.

(i) Write a balanced equation for this reaction.

.....

(ii) Explain why the catalyst should be in the form of a powder supported on a mesh.

.....

.....

.....

[3]

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A5 Electrolysis is the decomposition of a liquid by the passage of an electrical current.

(a) Aqueous copper(II) sulphate contains the following ions, Cu^{2+}, H^+, OH^- and SO_4^{2-}. Aqueous copper(II) sulphate can be electrolysed using inert electrodes.

The electrode reactions are represented below.

cathode $Cu^{2+} + 2e^- \longrightarrow Cu$
anode $4OH^- \longrightarrow O_2 + 2H_2O + 4e^-$

(i) Explain why copper, **not** hydrogen, is formed at the cathode.

.....

.....

(ii) Explain why the formation of oxygen at the anode is an example of oxidation.

.....

.....

(iii) The electrolysis of aqueous copper(II) sulphate using copper electrodes has a different anode reaction.
Give the equation for the electrode reaction at the anode.

.....
[3]

(b) Molten lead(II) bromide decomposes when an electric current is passed through it.

(i) Explain why solid lead(II) bromide will not conduct electricity but molten lead(II) bromide will.

.....

.....

.....

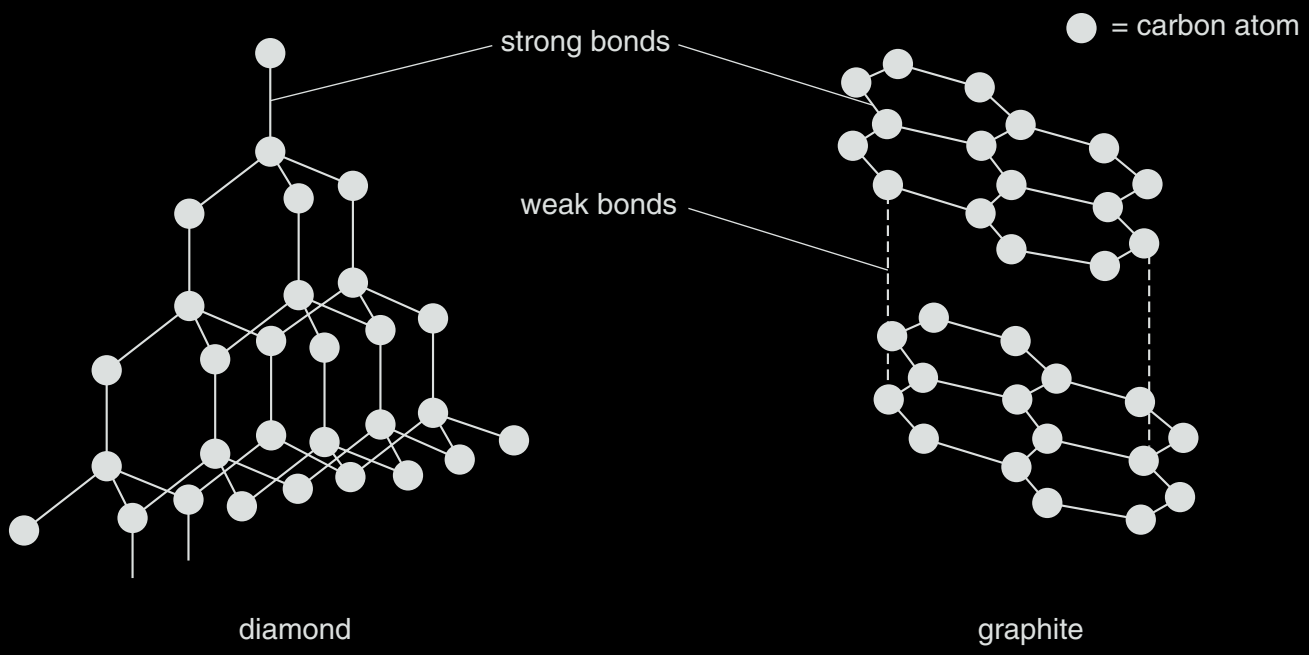
(ii) Construct the equations for the two electrode reactions.

cathode .....

anode .....
[4]

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A6 The structures of diamond and graphite are drawn below.



(a) Name the type of strong bond shown on the diagram.

.....[1]

(b) Diamond has a melting point of about 3700 °C and graphite has a melting point of about 3300 °C.

(i) Explain why both diamond and graphite have very high melting points.

.....

.....

.....

(ii) Suggest why the melting point of graphite is lower than that of diamond.

.....

.....
[3]

(c) Compare the electrical conductivity of diamond and graphite.
Explain your answer.

.....

.....

.....[2]

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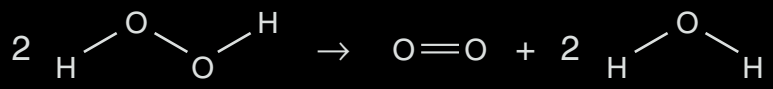
Section B

Answer **three** questions from this section.

B7 Aqueous hydrogen peroxide is used to sterilise contact lenses.

At room temperature aqueous hydrogen peroxide decomposes very slowly to form water and oxygen.

The decomposition can be represented by the equation below.



$\Delta H = -206\,\text{kJ/mol}$

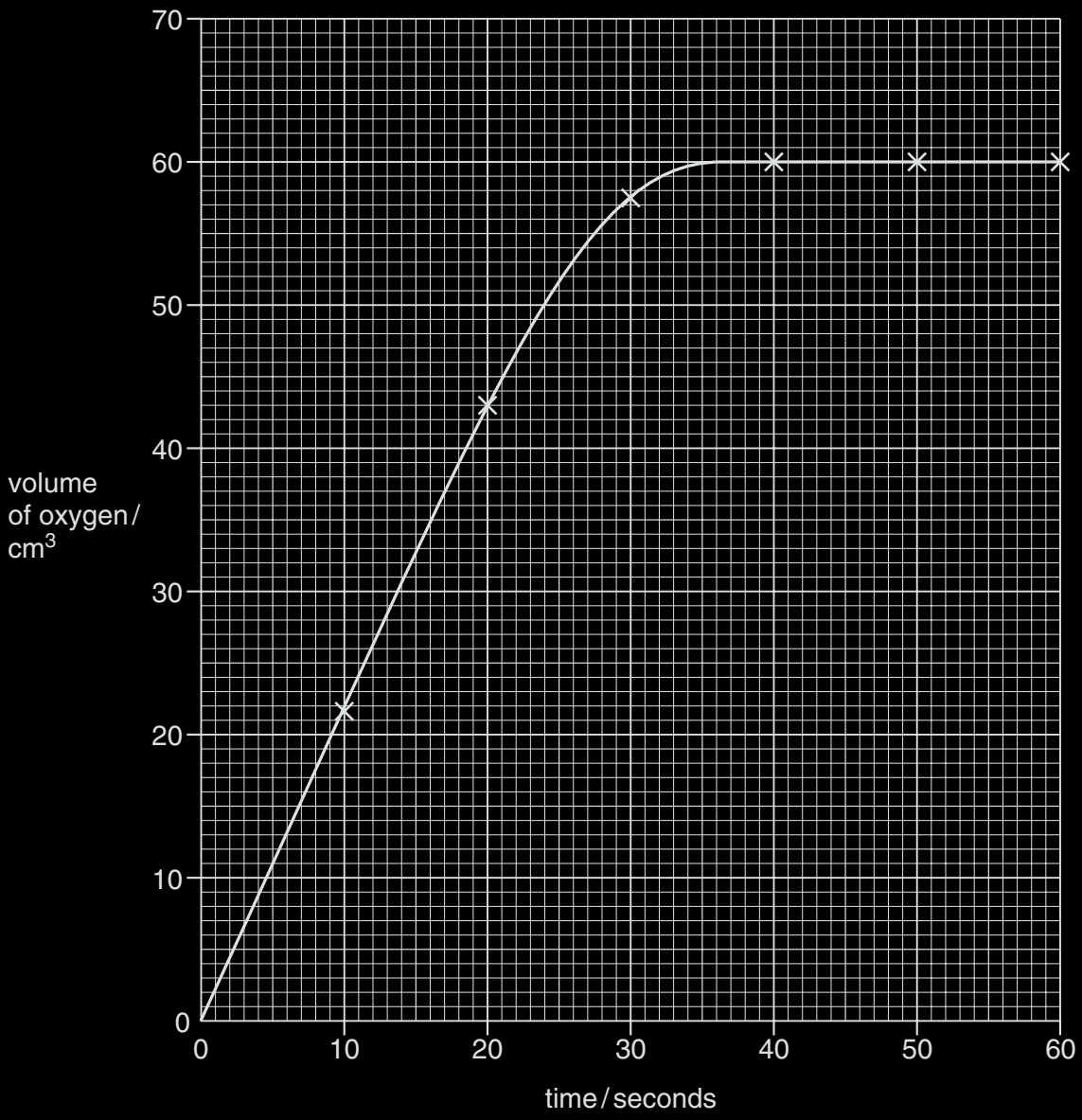
(a) Explain why this reaction is exothermic in terms of the energy changes that take place during bond breaking and bond making. [2]

(b) Draw the energy profile diagram for the decomposition of hydrogen peroxide.
Label on the diagram the activation energy and the enthalpy change. [3]

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(c) Manganese(IV) oxide catalyses the decomposition of aqueous hydrogen peroxide. In an experiment 50.0 cm^3 of aqueous hydrogen peroxide was mixed with 0.50 g of manganese(IV) oxide. The total volume of oxygen formed was measured every 10 seconds.

The results of the experiment are shown in the graph.



(i) After how many seconds did the decomposition of hydrogen peroxide finish?

(ii) How many moles of oxygen were produced at the end of the decomposition? [At room temperature and pressure one mole of oxygen occupies 24000 cm^3.]

(iii) Use your answer to **(ii)** to calculate the concentration, in mol/dm^3, of the 50.0 cm^3 of aqueous hydrogen peroxide used in the experiment.

[5]

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B8 Nickel is a transition element. It is manufactured in a four-stage process from nickel(II) sulphide, NiS.

- Stage 1 – nickel(II) sulphide is heated in air to form nickel(II) oxide and sulphur dioxide.
- Stage 2 – nickel(II) oxide is heated with carbon to give impure nickel.
- Stage 3 – impure nickel is reacted with carbon monoxide to make nickel tetracarbonyl, $Ni(CO)_4$.
- Stage 4 – nickel tetracarbonyl is decomposed to give pure nickel.

(a) **(i)** Construct the balanced equation for the reaction in stage 1.

(ii) Calculate the mass of sulphur dioxide that is formed when 182 kg of nickel sulphide is heated in air. [3]

(b) Nickel tetracarbonyl is a liquid with a boiling point of 43 °C.
Suggest, with a reason, the type of bonding in nickel tetracarbonyl. [2]

(c) Suggest **one** possible environmental consequence of the manufacture of nickel. [1]

(d) Give an example of the use of nickel as a catalyst. [1]

(e) In an experiment, small amounts of three metals were added to three aqueous metal nitrate solutions.
The results are shown in the table.

	aqueous zinc nitrate $Zn(NO_3)_2$	aqueous nickel(II) nitrate, $Ni(NO_3)_2$	aqueous copper(II) nitrate, $Cu(NO_3)_2$
zinc	no reaction	green solution went colourless and zinc coated with a silver solid	blue solution went colourless and zinc coated with a pink solid
nickel		no reaction	
copper	no reaction	no reaction	no reaction

Predict the observations when nickel is added to separate solutions of zinc nitrate and copper(II) nitrate.
Write an ionic equation for **one** of the reactions that takes place. [3]

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B9 Ethene is an important starting material for the production of chemicals such as ethanol, ethanoic acid and ethane-1,2-diol. Ethene, C_2H_4, is manufactured by the cracking of long chain hydrocarbons such as dodecane, $C_{12}H_{26}$.

(a) Construct an equation to show the cracking of dodecane to make ethene. [1]

(b) Draw a 'dot and cross' diagram for ethene. You only need to draw the valence (outer shell) electrons. [1]

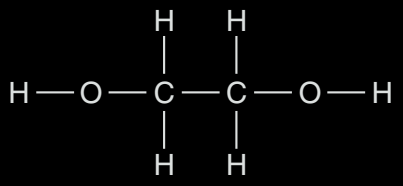
(c) Ethene can also be converted into a compound that contains carbon, hydrogen and oxygen. A sample of the compound was analysed and found to contain 0.72 g of carbon, 0.18 g of hydrogen and 0.96 g of oxygen.
Show that the empirical formula of the compound is CH_3O. [3]

(d) Describe how ethene can be converted industrially into ethanol. [2]

(e) Ethanol reacts with hot acidified potassium dichromate(VI) to form ethanoic acid.

(i) Describe the colour change that occurs during this reaction and draw the structure of ethanoic acid.

(ii) Ethane-1,2-diol has the structure drawn below.



Suggest the structure of the product of the reaction between ethane-1,2-diol and hot acidified potassium dichromate(VI).

[3]

B10 The table below shows some of the ores of iron.

ore	formula
haematite	Fe_2O_3
magnetite	Fe_3O_4
siderite	$FeCO_3$

(a) Which ore in the table contains the greatest percentage by mass of iron? Explain your answer. [2]

(b) Give the equations for the **redox** reactions taking place in the extraction of iron from haematite.
In each case state which substance is oxidised and which is reduced. [4]

(c) Iron is malleable. Describe how this property can be explained in terms of its structure. [2]

(d) State and explain how the properties of iron can be changed by the addition of carbon. [2]

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DATA SHEET
The Periodic Table of the Elements

Group I	Group II										Group III	Group IV	Group V	Group VI	Group VII	Group 0
							1 **H** Hydrogen 1									4 **He** Helium 2
7 **Li** Lithium 3	9 **Be** Beryllium 4										11 **B** Boron 5	12 **C** Carbon 6	14 **N** Nitrogen 7	16 **O** Oxygen 8	19 **F** Fluorine 9	20 **Ne** Neon 10
23 **Na** Sodium 11	24 **Mg** Magnesium 12										27 **Al** Aluminium 13	28 **Si** Silicon 14	31 **P** Phosphorus 15	32 **S** Sulphur 16	35.5 **Cl** Chlorine 17	40 **Ar** Argon 18
39 **K** Potassium 19	40 **Ca** Calcium 20	45 **Sc** Scandium 21	48 **Ti** Titanium 22	51 **V** Vanadium 23	52 **Cr** Chromium 24	55 **Mn** Manganese 25	56 **Fe** Iron 26	59 **Co** Cobalt 27	59 **Ni** Nickel 28	64 **Cu** Copper 29	65 **Zn** Zinc 30	70 **Ga** Gallium 31	73 **Ge** Germanium 32	75 **As** Arsenic 33	79 **Se** Selenium 34	80 **Br** Bromine 35	84 **Kr** Krypton 36
85 **Rb** Rubidium 37	88 **Sr** Strontium 38	89 **Y** Yttrium 39	91 **Zr** Zirconium 40	93 **Nb** Niobium 41	96 **Mo** Molybdenum 42	**Tc** Technetium 43	101 **Ru** Ruthenium 44	103 **Rh** Rhodium 45	106 **Pd** Palladium 46	108 **Ag** Silver 47	112 **Cd** Cadmium 48	115 **In** Indium 49	119 **Sn** Tin 50	122 **Sb** Antimony 51	128 **Te** Tellurium 52	127 **I** Iodine 53	131 **Xe** Xenon 54
133 **Cs** Caesium 55	137 **Ba** Barium 56	139 **La** Lanthanum 57 *	178 **Hf** Hafnium 72	181 **Ta** Tantalum 73	184 **W** Tungsten 74	186 **Re** Rhenium 75	190 **Os** Osmium 76	192 **Ir** Iridium 77	195 **Pt** Platinum 78	197 **Au** Gold 79	201 **Hg** Mercury 80	204 **Tl** Thallium 81	207 **Pb** Lead 82	209 **Bi** Bismuth 83	**Po** Polonium 84	**At** Astatine 85	**Rn** Radon 86
Fr Francium 87	226 **Ra** Radium 88	227 **Ac** Actinium 89 †															

140 **Ce** Cerium 58	141 **Pr** Praseodymium 59	144 **Nd** Neodymium 60	**Pm** Promethium 61	150 **Sm** Samarium 62	152 **Eu** Europium 63	157 **Gd** Gadolinium 64	159 **Tb** Terbium 65	162 **Dy** Dysprosium 66	165 **Ho** Holmium 67	167 **Er** Erbium 68	169 **Tm** Thulium 69	173 **Yb** Ytterbium 70	175 **Lu** Lutetium 71
232 **Th** Thorium 90	**Pa** Protactinium 91	238 **U** Uranium 92	**Np** Neptunium 93	**Pu** Plutonium 94	**Am** Americium 95	**Cm** Curium 96	**Bk** Berkelium 97	**Cf** Californium 98	**Es** Einsteinium 99	**Fm** Fermium 100	**Md** Mendelevium 101	**No** Nobelium 102	**Lr** Lawrencium 103

*58-71 Lanthanoid series
†90-103 Actinoid series

Key

a
X
b

a = relative atomic mass
X = atomic symbol
b = proton (atomic) number

The volume of one mole of any gas is 24 dm^3 at room temperature and pressure (r.t.p.).